RHEL 9 vs SUSE Linux Enterprise Server (SLES 15) — Detailed Comparison

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1. Introduction

This document provides a detailed technical and operational comparison between **Red Hat Enterprise Linux 9 (RHEL 9)** and **SUSE Linux Enterprise Server 15 (SLES 15, currently SP6)**, two leading enterprise-grade Linux distributions used heavily in datacenter, SAP, and mission-critical environments. It is intended for system administrators, DevOps engineers, and architects evaluating which OS best fits their workload, compliance, budget, and support requirements.

2. Overview & Lineage

	RHEL 9	SUSE Linux Enterprise Server 15
Vendor	Red Hat (IBM subsidiary)	SUSE Software Solutions (independent, EQT-backed)
Upstream/community lineage	Built from CentOS Stream 9 → Fedora origins	Built from openSUSE Leap / shares core with openSUSE Tumbleweed
First released	May 17, 2022	SLES 15 base: 2018; current Service Pack: SP6 (2024)
Versioning model	Major.Minor point releases (9.0 → 9.6)	Service Pack model (15 SP1 ... SP6) under one SLES 15 "major"
Philosophy	Stability, long-term binary compatibility, certification-first	Modular packaging, built-in rollback safety, strong SAP/mainframe focus
Governance	Corporate (Red Hat), community input via Fedora/CentOS Stream	Corporate (SUSE), community input via openSUSE project

3. Release & Support Lifecycle

	RHEL 9	SLES 15
General support	5 years → ends May 2027	Each Service Pack: ~2–3 years general support; base SLES 15 line runs 2018–2028
Extended support	Phase 2 to May 2029; ELS to May 2032	Long Term Service Pack Support (LTSS) extends each SP by up to 3 extra years
Update model	Point releases (9.0 → 9.6...)	Rolling Service Packs within SLES 15 (SP1...SP6), continuously patched between SPs
In-place major upgrade	`leapp` (RHEL 8→9, 9→10)	`zypper migration` (SP-to-SP), SUSE also supports in-place major version upgrades
Atomic/transactional updates	Not standard on RHEL 9 (available conceptually via rpm-ostree on Fedora CoreOS/OpenShift)	**`transactional-update`** native on SLE Micro / can be enabled on SLES — atomic, reboot-based rollback via Btrfs snapshots

4. Supported Architectures

Architecture	RHEL 9	SLES 15
x86_64	✅	✅
aarch64 (ARM64)	✅	✅
ppc64le (IBM Power)	✅	✅
s390x (IBM Z/mainframe)	✅	✅ (SUSE has historically very deep IBM Z/mainframe integration)
RISC-V	❌	Experimental via openSUSE, not SLES-certified

5. Installation & Administration Tools

	RHEL 9	SLES 15
Installer	Anaconda (graphical/text/kickstart)	YaST installer (graphical/text, AutoYaST for automation)
Automated install	Kickstart (`.ks` files)	AutoYaST (XML profiles)
Post-install admin GUI/TUI	Cockpit (optional web UI)	**YaST Control Center** — unified GUI/ncurses tool for nearly all system config (network, storage, users, services, firewall) — a signature SUSE feature
Default partition layout	LVM + XFS on `/`	LVM + **Btrfs** on `/` with Snapper pre-configured for automatic snapshots
Fleet/config management (vendor)	Red Hat Satellite	SUSE Manager

6. Package Management (Deep Dive)

	RHEL 9	SLES 15
Low-level tool	`rpm`	`rpm` (same package format — both use RPM!)
High-level tool	`dnf`	`zypper`
Repo config location	`/etc/yum.repos.d/*.repo`	`/etc/zypp/repos.d/*.repo`
Dependency resolver	libdnf/libsolv	libzypp (also libsolv-based, same solver lineage as dnf)
Modular/versioned software	AppStream module streams	Modules via SUSE Customer Center module system (e.g., Python, Node.js, Web Scripting modules) — conceptually similar
Universal/sandboxed apps	Flatpak (opt-in)	Flatpak (opt-in), Snap also supported but far less emphasized than on Ubuntu
Patch bundling	Errata/advisories via `dnf updateinfo`	SUSE patches are delivered as atomic "patch" units via `zypper patch` (groups related RPM updates + metadata into one logical patch)
Rollback after update	No native transactional rollback (manual downgrade only)	**Snapper + Btrfs** — automatic pre/post snapshot on every zypper transaction, instant rollback via `snapper rollback` or boot menu

> **Key differentiator:** Both RHEL and SUSE use the **RPM** package format (unlike Ubuntu's DEB), making cross-skill RPM knowledge largely portable — but SUSE's Btrfs+Snapper combo gives it a safety net RHEL/dnf doesn't have by default: botched updates can be rolled back by booting a prior snapshot.

7. Filesystem & Storage

	RHEL 9	SLES 15
Default root filesystem	**XFS**	**Btrfs** (with subvolumes for /home, /var, /opt, etc.)
Built-in snapshot tooling	LVM snapshots (manual), Btrfs tech-preview only	**Snapper** — fully integrated, automatic snapshots on package install/update/config change
Volume mgmt abstraction	Stratis	LVM thin provisioning; Btrfs subvolumes natively replace much of the need for a separate volume manager
Alternative FS support	ext4, Btrfs (unsupported), Stratis	XFS, ext4 also fully supported as alternates for data volumes/databases
Recommended FS for databases	XFS (default, performs well for large sequential I/O)	XFS commonly recommended for `/var/lib/mysql`, `/var/lib/pgsql` etc. even though Btrfs is root default — SUSE documentation explicitly suggests non-Btrfs for DB data directories

8. Kernel, Toolchain & Boot Process

	RHEL 9	SLES 15 (SP6)
Kernel (approx.)	5.14, heavily backported by Red Hat	6.4 kernel (SP6), heavily backported by SUSE — SUSE historically ships newer kernel bases than RHEL within similar timeframe
Kernel live patching	Red Hat kernel live patching (subscription add-on, limited scope)	**KLP (kGraft-based) via SUSE Manager/kernel-livepatch** — long-standing, mature live-patching heritage (SUSE pioneered kGraft)
Bootloader	GRUB2 (BLS)	GRUB2, tightly integrated with Btrfs snapshot boot entries (boot into any snapshot from GRUB menu)
initramfs tool	`dracut`	`dracut` (shared tooling with RHEL/Fedora lineage)
Default compiler	GCC 11 (gcc-toolset for newer)	GCC 12/13 depending on SP
Python (system default)	3.9 (3.11/3.12 via module)	3.6 base system Python (SLES is conservative here), newer via modules

9. Security

	RHEL 9	SLES 15
Mandatory Access Control (default)	**SELinux** — enforcing by default	**AppArmor** — enabled by default (SELinux is available/supported as an alternative but not default)
Firewall front-end	`firewalld`	`firewalld` (SLES 15 adopted firewalld as default, replacing the older SuSEfirewall2)
System-wide crypto policy	`update-crypto-policies` (DEFAULT/LEGACY/FUTURE/FIPS)	Similar concept via `update-crypto-policies` is not native; crypto hardening typically managed per-service, though FIPS mode is supported
FIPS 140-3 validated modules	Yes, first-class	Yes — SLES has long-standing FIPS 140-2/140-3 certified crypto modules, important for government/finance customers
Gov/compliance certs	Common Criteria, FIPS, DISA STIG	Common Criteria, FIPS — SUSE also maintains strong certification history, particularly in European government/telco sectors
Auditing	`auditd`, integrated with SELinux	`auditd` available, integrated with AppArmor logging via `journalctl` / `dmesg`

10. User & Group Management

	RHEL 9	SLES 15
Admin group	`wheel`	`wheel` or direct root login tradition (historically SUSE allowed root login more readily; modern SLES defaults mirror RHEL's sudo-first model)
User management tool	`useradd` / `usermod`, Cockpit "Accounts"	`useradd` / `usermod`, or via YaST "Users and Groups" module
Central identity integration	SSSD, tightly integrated with FreeIPA/AD	SSSD, also integrates with Samba/AD; YaST provides a GUI wizard for domain join

11. Networking

	RHEL 9	SLES 15
Primary network manager	NetworkManager (`nmcli` , `nmtui`)	**wicked** (SUSE's own network management daemon) by default on SLES, though NetworkManager is selectable, especially for desktop/laptop use
Config file location	`/etc/NetworkManager/system-connections/`	`/etc/sysconfig/network/ifcfg-*` (wicked) or NetworkManager connection files if selected
GUI network config	`nm-connection-editor`	YaST → Network Settings module
Hostname tool	`hostnamectl`	`hostnamectl`

12. Logging & Monitoring

	RHEL 9	SLES 15
Journal	`systemd-journald` / `journalctl`	`systemd-journald` / `journalctl`
Traditional syslog	`rsyslog` → `/var/log/messages`	`rsyslog` → `/var/log/messages` (same convention as RHEL)
Log rotation	`logrotate`	`logrotate`
Built-in diagnostics	`sos report` , Red Hat Insights	`supportconfig` (SUSE's equivalent of sosreport, feeds directly into SUSE support ticketing)
Fleet monitoring/management	Red Hat Satellite, Insights	SUSE Manager (patch/config/subscription management across fleets)

13. Process & Service Management

	RHEL 9	SLES 15
Init/service manager	systemd	systemd
Custom unit file location	`/etc/systemd/system/`	`/etc/systemd/system/`
Service enable/start	`systemctl enable --now <svc>`	`systemctl enable --now <svc>`

Service management is effectively identical, since both use systemd — administrators moving between the two will find almost no learning curve here.

14. Containers & Virtualization

	RHEL 9	SLES 15
Default container engine	**Podman** (daemonless, rootless-first)	**Podman** also the recommended engine on recent SLES/SLE Micro; Docker available via containers module
First-party orchestration	OpenShift (commercial)	SUSE Rancher / RKE2 — SUSE acquired Rancher, making it a major Kubernetes distribution vendor
Minimal/immutable OS variant	Fedora CoreOS / RHEL CoreOS (OpenShift nodes)	**SLE Micro** — immutable, transactional-update based OS purpose-built for containers/edge, tightly paired with Rancher/K3s
Hypervisor	KVM/QEMU via libvirt, oVirt/RHV at scale	KVM/QEMU via libvirt; Xen also historically supported on SUSE (legacy, being phased toward KVM)

15. High Availability & Clustering

	RHEL 9	SLES 15
Cluster stack	Pacemaker + Corosync, Red Hat supported	**SUSE Linux Enterprise High Availability Extension** — Pacemaker + Corosync, arguably SUSE's flagship enterprise strength, with deep SAP HANA System Replication cluster integration
Shared storage clustering	GFS2	OCFS2, also supports GFS2-equivalent cluster filesystems
SAP-specific HA tooling	Supported, but SAP ecosystem leans toward SUSE for HANA HA	**SAPHanaSR** resource agents — purpose-built, SUSE-maintained, considered the reference implementation for SAP HANA HA clustering

16. Performance Tuning

	RHEL 9	SLES 15
Tuning framework	`tuned` with profiles	`tuned` also available and used, plus SUSE-specific SAP tuning packages (`saptune`) that auto-apply SAP-recommended kernel/sysctl settings
SAP-specific tuning	Manual application of SAP notes via `tuned` profiles	`saptune` — SUSE's dedicated, widely regarded as best-in-class tool for applying/validating SAP OS tuning notes automatically

17. Licensing & Cost

	RHEL 9	SLES 15
Base OS acquisition	Free download; updates require active subscription (free dev tier capped at 16 systems)	Free download/evaluation (60-day trial); production updates require a subscription registered via SUSE Customer Center (SCC)
Support tiers	Self-Support, Standard, Premium (24/7)	Standard and Priority support tiers, similarly priced per system/socket-pair
Cloud marketplace pricing	Often an hourly RHEL surcharge	SLES images also typically carry an hourly surcharge on public clouds, though SLES for SAP often bundled favorably with SAP cloud deployments
Subscription management tool	`subscription-manager` (Red Hat Subscription Management / Satellite)	`SUSEConnect` (ties to SUSE Customer Center or local SUSE Manager/RMT server)

18. Ecosystem & Vendor Support (SAP Focus)

	RHEL 9	SLES 15
SAP HANA certification	Certified, widely used	**Historically the primary/preferred OS for SAP HANA** — SAP itself has used SUSE internally and SLES for SAP Applications is a dedicated, pre-tuned product edition
Mainframe (IBM Z) focus	Strong	Very strong — SUSE has one of the longest-running IBM Z Linux partnerships in the industry
Hardware vendor certification	Deep OEM matrix (Dell, HPE, Lenovo, Cisco)	Also deep, especially where SAP workloads are involved (HPE, Lenovo, Fujitsu often co-certify SAP+SLES reference architectures)
Kubernetes/cloud-native ecosystem	OpenShift-centric	Rancher/RKE2-centric (SUSE owns Rancher, a major CNCF ecosystem player)
Community size	Large; Red Hat Customer Portal (subscription-gated KB)	Smaller than RHEL/Ubuntu overall, but strong openSUSE community and SUSE-specific forums

19. Pros & Cons Summary

RHEL 9

Pros:

- Largest enterprise Linux ecosystem/market share — most third-party documentation, tooling, and ISV certifications overall.
- SELinux provides very granular, label-based mandatory access control.
- `leapp` gives a mature in-place major-version upgrade path.
- OpenShift gives a strong first-party enterprise Kubernetes story.

Cons:

- No native transactional rollback for the base OS/updates (XFS has no built-in snapshot+rollback workflow like Btrfs/Snapper).
- Subscription required for ongoing patching.
- Less specialized SAP HANA tooling heritage compared to SUSE's `saptune`/SAPHanaSR.

SUSE Linux Enterprise Server 15

Pros:

- **Btrfs + Snapper** gives built-in, automatic snapshot/rollback for nearly every system change — a major operational safety net RHEL lacks by default.
- **Best-in-class SAP HANA tooling** (`saptune`, SAPHanaSR) — the de facto reference OS for SAP HANA HA deployments.
- Mature kernel live-patching heritage (kGraft) for near-zero-downtime kernel security patching.
- YaST provides a genuinely unified single tool for most system administration tasks — easier onboarding for less CLI-heavy admins.
- Owns Rancher — strong, independent Kubernetes/cloud-native story (RKE2, K3s) not tied to a single orchestration stack.

Cons:

- Smaller overall market share/community compared to RHEL — fewer generic how-to guides and third-party ISV certifications outside SAP/mainframe niches.
- AppArmor (default) is generally considered less granular than SELinux for complex MAC policies, though it's simpler to author.
- wicked (default network stack) is less familiar to admins coming from RHEL/Ubuntu/NetworkManager backgrounds.
- Running databases directly on default Btrfs root without redirecting data directories to XFS/ext4 can introduce performance/fragmentation concerns if not configured carefully.

20. Decision Matrix — Which to Choose

Scenario	Recommended OS	Why
SAP HANA production deployment, especially HA	**SLES 15**	`saptune` + SAPHanaSR are the industry reference tooling for SAP HANA
Need automatic rollback safety net for patching	**SLES 15**	Btrfs + Snapper gives built-in pre/post snapshot rollback
Broadest third-party ISV certification coverage (non-SAP)	**RHEL 9**	Largest overall enterprise Linux market share and certification base
Fine-grained, label-based MAC security policy requirements	**RHEL 9**	SELinux offers more granular control than AppArmor
IBM Z mainframe-heavy environment	**SLES 15**	SUSE has one of the industry's longest-running Z Linux partnerships
Kubernetes via Rancher/RKE2/K3s ecosystem	**SLES 15**	SUSE owns Rancher — tightly integrated, vendor-unified stack
OpenShift-based container platform	**RHEL 9**	OpenShift is Red Hat's first-party enterprise K8s platform
Admins who prefer a single unified GUI/TUI for all system config	**SLES 15**	YaST Control Center consolidates most admin tasks in one tool

21. Migration Considerations

- **Packages**: Both use RPM, but packages are **not directly cross-installable** due to differing dependency naming, library versions, and repo metadata — rebuild/re-source packages for the target distro.
- **Security policy**: SELinux policies and AppArmor profiles are not portable — MAC policy must be redesigned when switching.
- **Network stack**: Moving to SLES may require re-learning `wicked` / `ifcfg-*` conventions if coming from a pure NetworkManager background (or simply switch SLES to use NetworkManager instead).
- **Root filesystem**: Migrating to SLES's Btrfs-by-default layout means planning Snapper snapshot retention policies (they consume disk space over time) and deciding which data directories (databases, logs) should be redirected to non-Btrfs subvolumes/filesystems.
- **Automation**: Ansible/Terraform playbooks referencing `dnf` vs `zypper` module names, or OS-specific facts (`ansible_os_family`: RedHat vs Suse), need conditional branches.
- **Validate SAP/ISV certification** explicitly for the target OS — this is often the deciding factor in RHEL↔SUSE migrations given how SAP-centric the SUSE value proposition is.

22. Directory & Config File Locations

Purpose	RHEL 9	SLES 15
Package repo config	`/etc/yum.repos.d/`	`/etc/zypp/repos.d/`
Network config	`/etc/NetworkManager/system-connections/`	`/etc/sysconfig/network/ifcfg-*` (wicked)
Firewall rules	`/etc/firewalld/`	`/etc/firewalld/` (same, SLES 15 adopted firewalld)
Sudo rules	`/etc/sudoers` , `/etc/sudoers.d/`	`/etc/sudoers` , `/etc/sudoers.d/`
MAC policy	`/etc/selinux/`	`/etc/apparmor.d/`
systemd unit overrides	`/etc/systemd/system/`	`/etc/systemd/system/`
Snapshot config	N/A (LVM snapshots managed manually)	`/etc/snapper/configs/`
Subscription registration	`/etc/rhsm/`	`/etc/SUSEConnect` , `/etc/zypp/credentials.d/`
Default log directory	`/var/log/` (messages, secure, audit/)	`/var/log/` (messages, audit/)

23. Expanded Command Cheat Sheet

Package Management

Task	RHEL 9	SLES 15
Install package	`dnf install <pkg>`	`zypper install <pkg>`
Remove package	`dnf remove <pkg>`	`zypper remove <pkg>`
Update all	`dnf update`	`zypper update` (or `zypper patch` for official patches)
Search	`dnf search <term>`	`zypper search <term>`
List installed	`dnf list installed`	`zypper se --installed-only`
Show package info	`dnf info <pkg>`	`zypper info <pkg>`
List repos	`dnf repolist`	`zypper repos`
Clean cache	`dnf clean all`	`zypper clean`

Snapshots / Rollback (SLES-unique)

Task	RHEL 9	SLES 15
List snapshots	N/A (manual LVM snapshot listing via `lvs`)	`snapper list`
Create snapshot	N/A	`snapper create --description "before change"`
Rollback to snapshot	N/A	`snapper rollback <number>` then reboot

Service / Process Management

Task	RHEL 9	SLES 15
Start/stop/restart service	`systemctl start/stop/restart <svc>`	`systemctl start/stop/restart <svc>`
Enable at boot	`systemctl enable <svc>`	`systemctl enable <svc>`
View logs for service	`journalctl -u <svc>`	`journalctl -u <svc>`

Networking

Task	RHEL 9	SLES 15
Show IP config	`ip a` / `nmcli dev show`	`ip a` / `wicked show all`
Restart networking	`systemctl restart NetworkManager`	`systemctl restart wicked` (or `NetworkManager` if selected)
Edit interface config	via `nmcli` / `nmtui`	Edit `/etc/sysconfig/network/ifcfg-<iface>` or via YaST

Security

Task	RHEL 9	SLES 15
MAC status	`sestatus`	`aa-status`
Set MAC mode	`setenforce 0/1`	`aa-enforce` / `aa-complain <profile>`
Firewall allow port	`firewall-cmd --add-port=80/tcp --permanent && firewall-cmd --reload`	`firewall-cmd --add-port=80/tcp --permanent && firewall-cmd --reload`
Check open ports	`ss -tulpn`	`ss -tulpn`

System Info & Subscription

Task	RHEL 9	SLES 15
OS version	`cat /etc/redhat-release`	`cat /etc/os-release`
Kernel version	`uname -r`	`uname -r`
Subscription/support status	`subscription-manager status`	`SUSEConnect --status`
Apply SAP tuning (SLES-unique)	N/A (manual via `tuned`)	`saptune solution apply <SOLUTION>`

24. References

- Red Hat Enterprise Linux 9 Documentation — access.redhat.com/documentation
- Red Hat Product Life Cycle Dates — access.redhat.com/support/policy/updates/errata
- SUSE Linux Enterprise Server 15 Documentation — documentation.suse.com/sles/15-SP6
- SUSE Product Life Cycle — scc.suse.com / suse.com/lifecycle
- SAP on SUSE Linux Enterprise — suse.com/products/sles-for-sap

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